

Analysis on Diminos Store Delivery Time

Columns: order_id, order_placed_at, order_delivered_at

Number of records: ~15000

Number of missing values found: 0

Number of duplicate records found: 0

Summary Report: Based on a comprehensive analysis of 15,000 order records, the store is currently meeting the performance metric required by Diminos Corporate. The 95th percentile for order delivery time stands at 27.26 minutes, which is successfully below the 31-minute threshold. Consequently, the franchise license is secure under current operating conditions. However, the analysis revealed specific operational bottlenecks at 11:00 a.m., 4:00 p.m., and 1:00 AM where performance consistently fails to meet the target, posing a risk for future audits.

Performance Against Franchise Metrics: The primary objective of this analysis was to determine if the store's 95th percentile delivery time is less than 31 minutes.

Target: < 31 Minutes | Actual Performance: 27.26 Minutes | Status: PASSED

While the vast majority of orders are delivered efficiently, 3.71% of all orders (557 total) exceeded the 31-minute mark. Under the "31 Minutes or Free" promise, these orders represent a direct loss of revenue and inventory.

Operational Risk Analysis (Hourly Bottlenecks): By breaking down the 95th percentile performance by hour, the analysis uncovered critical inefficiencies that the daily average hides. The store specifically fails to meet the 31-minute requirement during three distinct time windows:

11:00 AM: 95th Percentile = 33.41 minutes

4:00 PM (16:00): 95th Percentile = 32.44 minutes

1:00 AM: 95th Percentile = 32.10 minutes

These specific hours were flagged as "risk hours" where the delivery time consistently exceeds the allowed threshold.

Outlier Analysis & Findings

1. Visual Detection: Initial visualization using a boxplot revealed the presence of extreme outliers in the dataset. The distribution was heavily skewed, with valid delivery times compressed into a small range while extreme values stretched the scale significantly.

2. Statistical Thresholds (IQR Method): To identify these outliers mathematically, we calculated the Interquartile Range (IQR):

25th Percentile (Q1): 15.27 minutes

75th Percentile (Q3): 17.28 minutes

Upper Bound ($Q3 + 1.5 * IQR$): ~20.29 minutes

Lower Bound ($Q1 - 1.5 * IQR$): ~12.27 minutes

Impact of Removal: Applying these strict statistical bounds filtered out a significant portion of the "long-tail" deliveries.

Original Dataset: 15,000 records

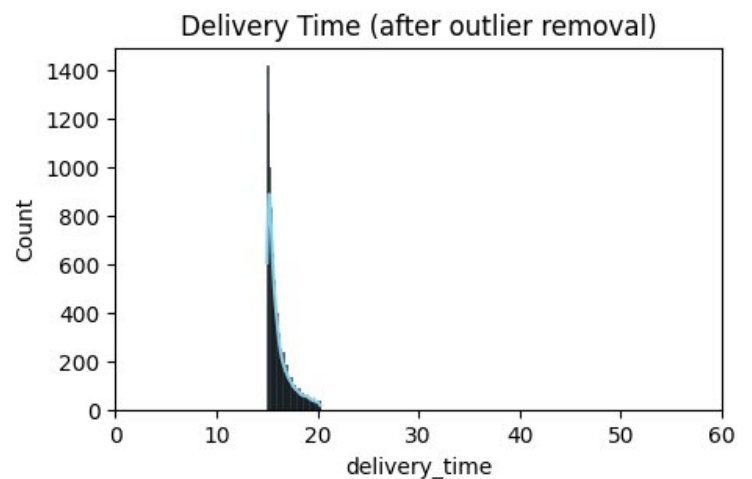
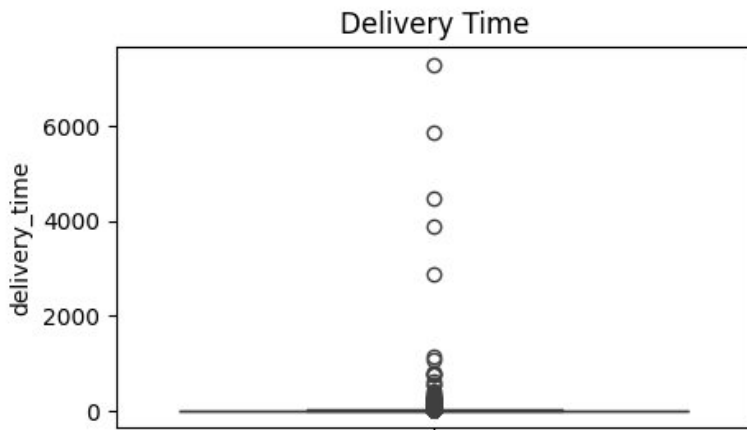
Records after Filtering: 13,266 records

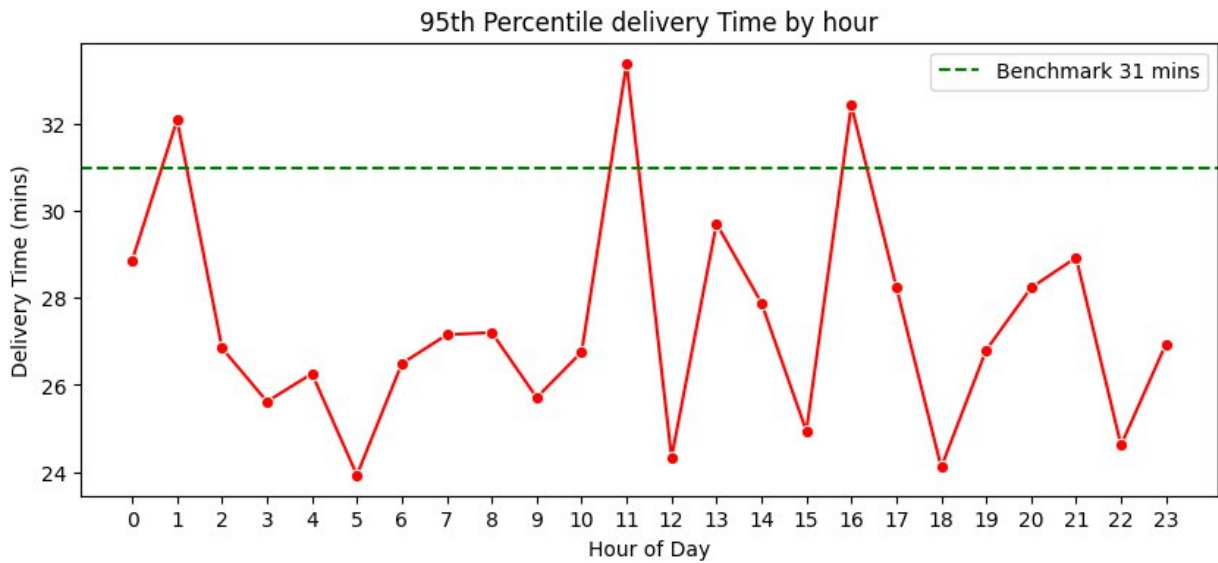
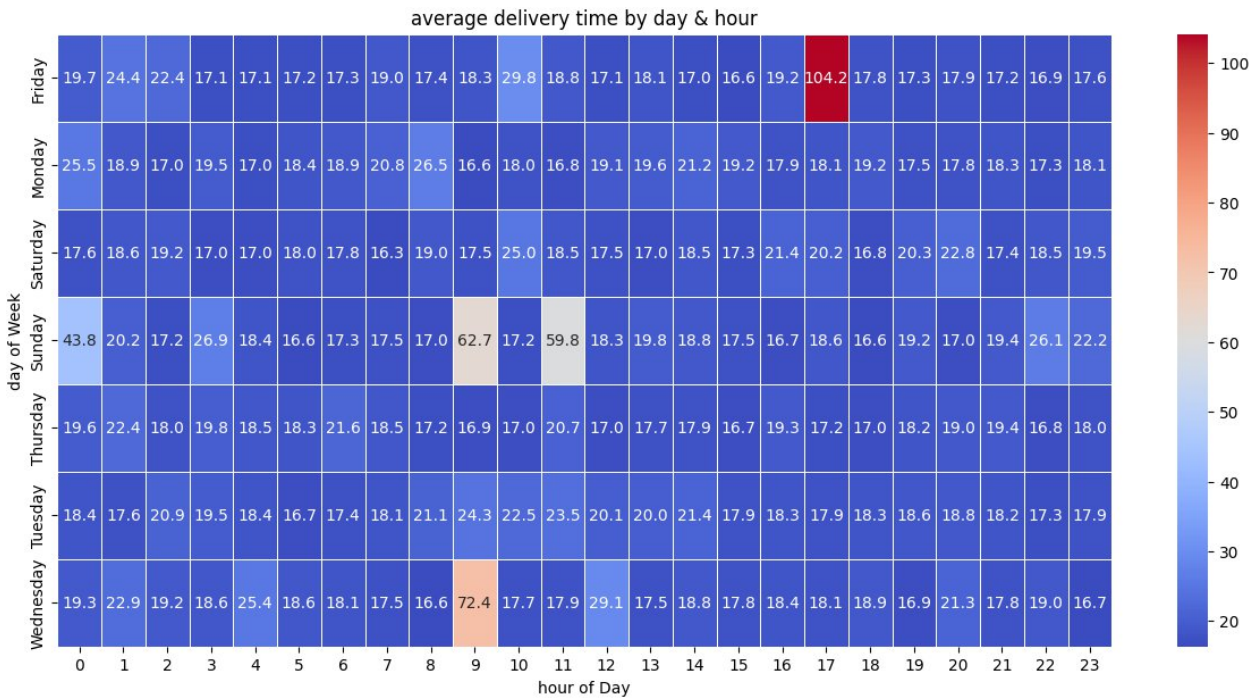
Result: Approximately 1,734 records fell outside the typical delivery window of 12–20 minutes. This confirms that while the core operation is highly consistent (around 15–17 minutes), there is a long tail of delayed orders that skew the average.

Strategic Recommendations: Based on the risks identified in the hourly analysis:

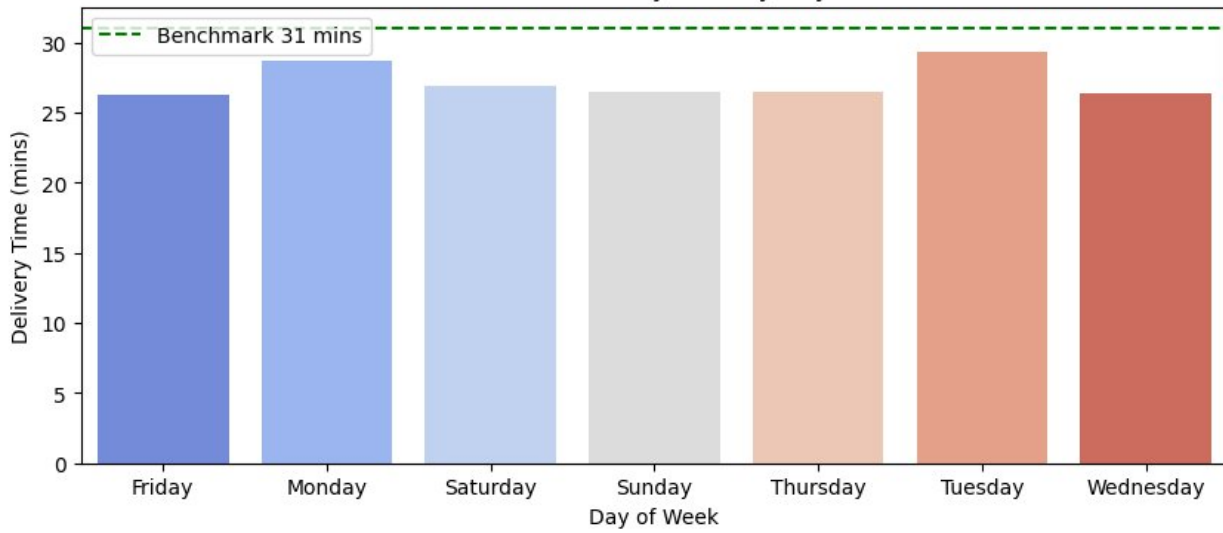
- Staffing Adjustments: Resources need to be reallocated to cover the high-risk periods of 11:00 AM and 4:00 PM to bring these specific hours under the 31-minute target.
- Shift Management: The 1:00 AM lag suggests a potential issue with late-night shift transitions or staffing levels that requires immediate investigation.
- Data Hygiene: A process must be enforced to prevent "multi-day" delivery records (e.g., the 7,299-minute outlier) by ensuring drivers close out orders in real-time.

Visualizations:

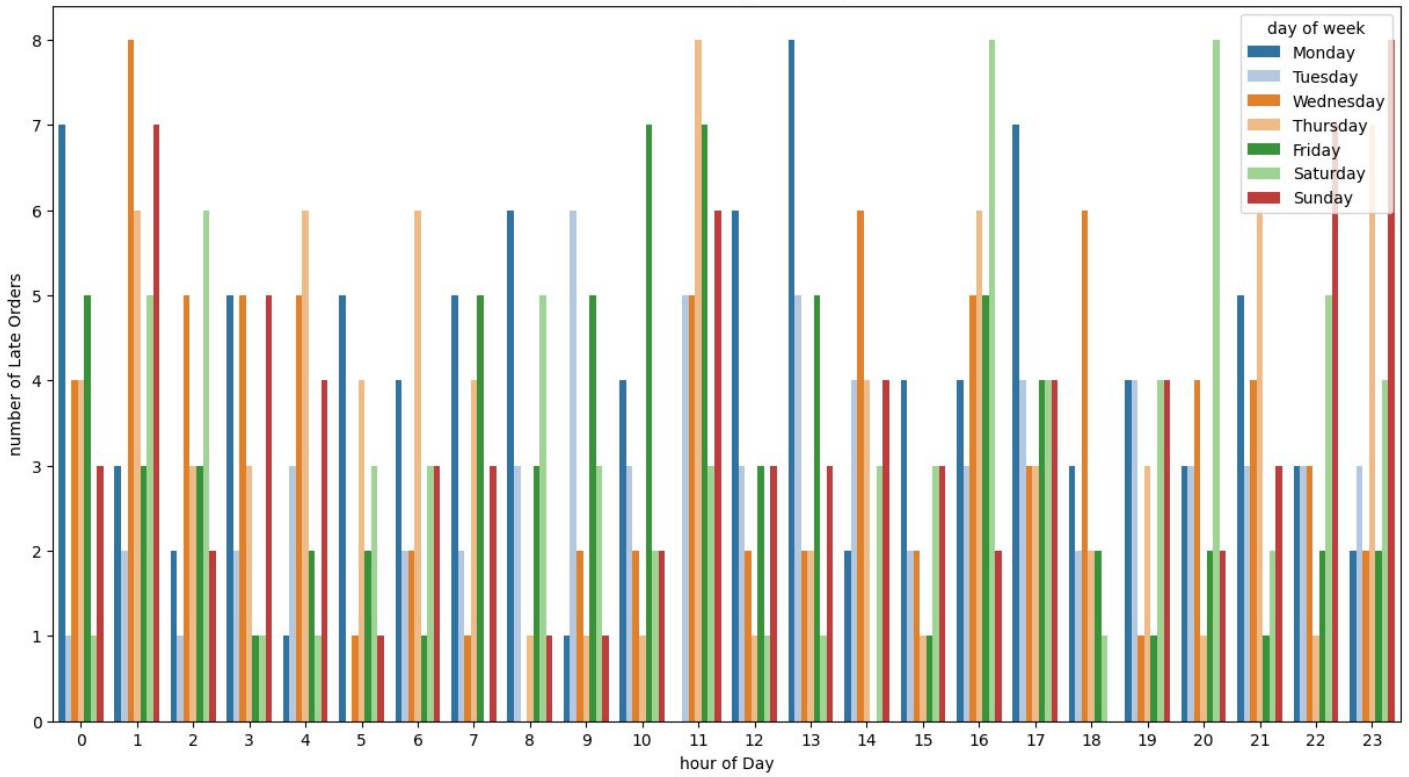




95th Percentile Delivery Time by Day of Week



late deliveries (>31 mins)



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